FLIP THE SWITCH · BEST BUSINESS WORKFLOW · DEMO GUIDE

AirGrid

A city's drone response team, run from team chat. Drones patrol a fictional city. AI agents for Drone Ops, Fire, Hospital and Police work each emergency together in Switch rooms on Discord, and the people on the team make every decision in the same chat.



The ops wall: the live 3D city, with two incidents running at once.

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SIMULATED AirGrid is a training and prototyping simulator. The district, the agency and every number on screen are simulated. No real drones, no real dispatch, no identification of people. The "thermal sensor" is simulated too.

Who's who

Channels (Switch rooms on Discord)

Channel	What happens there	Who's in it
`#ops`	Information only for incidents: one line when an incident starts, and the final report. This is also where the Watch Commander types fleet orders (`watch` , `dock` , `release`).	World Bot, Drone Ops, you
`#incident-1` , `#incident-2`	One incident each. The agents hand work to each other by @mention, and every decision for a person is asked here: the doctor's `transport` / `treat` and the lead's `leave` / `stay` . A second incident at the same time gets the other room.	World Bot, all 4 agents, you
`#fleet`	The whole fleet, live: every drone change about every 10 s, a fleet board every 2 minutes, an alert when a patrol drone goes home with no spare, and every order you gave. No @mentions, no replies.	World Bot only

The team

Member	Kind	Job
Drone Ops (`@dispatch`)	AI agent	Reports what the drones see, tags the right service, escorts the ambulance, relays requests. Answers "what's happening?" from the simulator.
Fire Station (`@fire-station`)	AI agent	Acknowledges fires and sends engines.
Hospital (`@hospital`)	AI agent	Sends ambulances and recommends treating on scene or taking the patient in.
Police (`@police`)	AI agent	Checks scenes for injuries, clears junctions and traffic jams.
World Bot (`@airgrid-worldbot`)	Plain code, no AI	A registered Switch client. Carries simulator events into the rooms and people's decisions back. It's the only member that can change the simulation, and only for a mapped person.

People and their roles

Roles live in `config/humans.yaml` . Each person is matched by their Switch sender id, so changing a display name can't fake a role. Solo, one person holds every role; with a team, give each role to someone different.

Role	Decides
Incident lead	`leave` or `stay` : whether the drone can leave a scene that looks clear.
Duty doctor	`transport` or `treat` : take the injured to the Hospital, or treat on scene.
Watch commander	Fleet orders: `watch` an area, `dock` or `release` a drone.

The principle behind all of it: **agents propose, people decide, the simulator enforces.** No agent can approve its own work, and time-critical work (launches, battery relief, landing slots, no-fly rules) is simulator code, so the city keeps running even if every agent goes offline.

The dashboard (the ops wall)

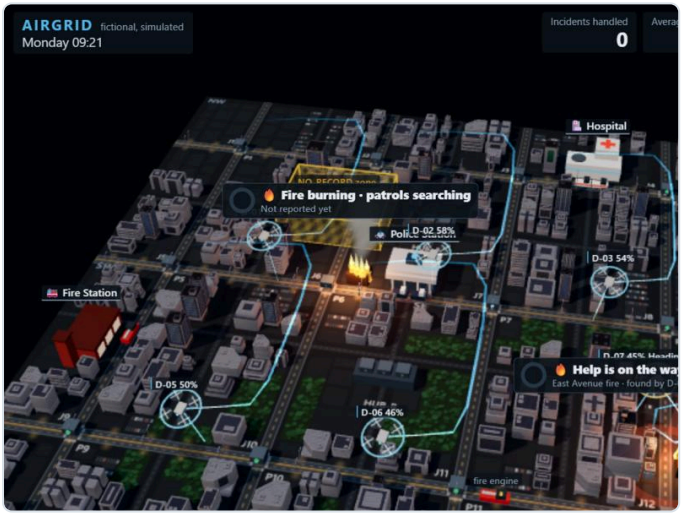
Open the URL that `scripts/run_game.sh --discord` prints. It includes a one-time operator key, so don't share it. The dashboard is **watch-only** for the drone monitor team. Decisions are made in Discord; the only inputs are the three simulate buttons.

What you see

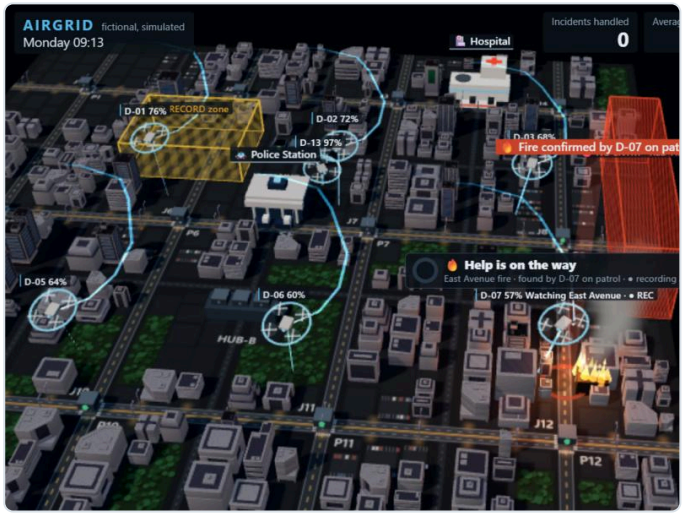
- 14 drones patrolling, each with its battery and job
- Fire Station, Hospital and Police Station, with vehicles that drive on the roads
- Progress rings at scenes: "Extinguishing 64%", "Treating patients 40%"
- A red no-entry airfield and a no-record zone, both enforced by the sim
- Top right: incidents handled, average response, drones up
- Bottom left: one plain line saying what's happening

The three buttons

🔥 **Simulate fire**	A reported fire at a city scene. The nearest drone diverts and the Fire Station responds.
💥 **Simulate accident**	A reported crash. Police check for injuries; with injuries, an ambulance goes and the doctor decides.
🎲 **Random fire**	A fire nobody reported. It burns on the map, and a patrol drone has to find it with its simulated thermal sensor. One at a time; the button says "searching…" until it's found.



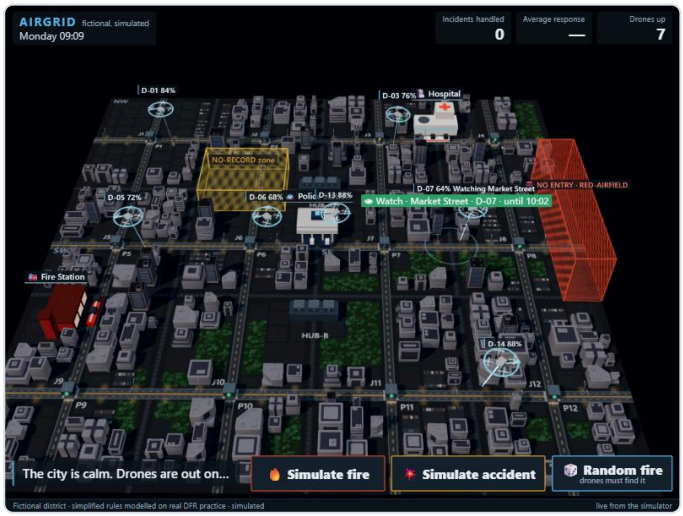
Random fire: burning, no alert beacon, "patrols searching". Agents and Discord know nothing yet.



A patrol drone confirms it ("Fire confirmed by D-07 on patrol") and the normal fire response starts.



Several incidents at once: each label says its stage and who found it.



A `watch market` order: the ring over Market Street and the drone "Watching Market Street".

Discord commands: the cheat sheet

Type them as a whole message, with no @mention. Upper or lower case both work. The World Bot answers in the same room; anything that isn't a command is just chat.

Decisions on an incident (in the incident's room)

Type	Who	What happens
`transport`	Duty doctor	The ambulance takes the patient to the Hospital, with a drone escorting it.
`treat`	Duty doctor	The crew treats on scene; no transport.
`leave`	Incident lead	The scene looks clear: the drone returns to patrol and the report is written.
`stay` · `stay 2 more minutes`	Incident lead	The drone keeps watching, then asks again with a clearer reading. The sim sets the hold time.
`leave market` · `stay ring`	Incident lead	With two calls open, add the place's first word to say which one.

Fleet orders (in `#ops`)

Type	Who	What happens
`watch market`	Watch commander	The nearest spare patrol drone flies loops over Market Street for 60 clock-minutes (about 2 real minutes). No recording: it isn't an incident. If its battery runs low, another drone takes over.
`watch ring road 30 min` · `increase surveillance on north road for 45 minutes`	Watch commander	The same, for a set time (5 to 120 clock-minutes).
`stop watching market`	Watch commander	Ends the watch; the drone goes back on patrol.
`dock D-4`	Watch commander	The drone flies to the nearest free dock and stays grounded. Use it for a fault report, or to charge drones before a busy hour. Refused while it's at an incident or escorting.
`release D-4`	Watch commander	The drone can fly again once charged.

Places: Central Crossing, East Avenue, Market Street, Mill Lane shops, North Road, Northgate Flats, Ring Road, South Depot. Any unique part of the name works (`market`, `mill`). Drones: D-01 to D-14 (`D-4` and `d04` both work).

Ask the agents (mention them)

Type	What happens
`@dispatch what's happening?`	Drone Ops reads the simulator and answers in one line, numbers exactly as printed.
`@dispatch ask police to stay until the tow arrives`	Drone Ops relays it: "Drone Ops → @police: Aadarsh asks: ...". Relaying changes nothing in the sim.

Switch's own commands

`!list-documents` (in `#ops`)	Shows the room's documents, including the **Flight authorisation log** the World Bot writes one entry to per incident.
`!list-agents` · `!agents-status`	Who's in the room and whether they're online.

What each channel looks like

`#ops`: start line, orders, report

airgrid-worldbot APP
💥 Ring Road accident. Crews are working in #incident-2.

Aadarsh
watch market
airgrid-worldbot
✅ D-07 watches Market Street until 10:02 (no recording: not an incident).

airgrid-worldbot
✅ Ring Road accident: closed. Report INC-2 (simulated)
Response: first eyes on in 84 s
Who went: police car 1, ambulance 1
Recording: 09:41 to 09:58, at the scene only
Timeline: …

`#fleet`: the whole fleet, live

🛰 Fleet: 7 in the air (4 on patrol, 3 at incidents) · 4 charging · 3 ready (kept for emergencies). Since 16:00: 14 went to charge, 13 took off. Lowest in the air: D-08 41%.

09:17 D-13 takes off for patrol (100%)
09:19 D-07 heads to dock P7 (44%)
09:22 D-07 lands at P7 and charges (40%)
09:36 D-06 charged at HUB-B, ready (80%)

📌 Fleet: D-02 heads to dock P3 and is grounded until released (92%).
⚠️ Fleet: D-14 heads to a dock at 35% from patrol route 5. No spare drone to relieve it: 3 stay ready for emergencies.

`#incident-2`: the work and the decisions

airgrid-worldbot
💥 Ring Road accident: Drone Ops has eyes on. @police you're needed.
police
🚓 Police: on the way to Ring Road, ETA 2 min.
airgrid-worldbot
🚑 Ring Road accident: 3 injured, @hospital ambulance needed.
🚑 Ring Road accident: Hospital: 3 injured, condition needs the Hospital. Recommend transport. @Aadarsh your call: `transport` or `treat`.
Aadarsh
transport
airgrid-worldbot
✅ Ring Road accident: Duty Doctor confirms transport to the Hospital.
🛰 Ring Road accident: Drone Ops (D-03) escorting the ambulance, route clear.
🛰 Ring Road accident: Drone Ops (D-03) → @police ambulance stuck near North Road, please clear it.
police
🚓 Police: North Road, on it.
airgrid-worldbot
🚓 Ring Road accident: Police cleared the junction near North Road.
🚑 Ring Road accident: Hospital has the patient. Delivered.
🤷 @Aadarsh Ring Road accident looks clear. Reply: `leave` or `stay`.
Aadarsh
leave
airgrid-worldbot
✅ Ring Road accident: heading back to patrol.

The wording follows the bot's templates; agent replies vary slightly from run to run.

Demo flows to run on camera

Screen layout: dashboard on the left (about 60%), Discord on the right. Total under 3 minutes; cut waiting time in the edit.

A · The accident, end to end (about 1:30) ★

1. Press **Simulate accident**. `#ops` gets one line naming the incident room. Switch to `#incident-1`.
2. Drone Ops has eyes on and tags `@police`; a police car drives there on the map.
3. "3 injured, @hospital ambulance needed." The ambulance drives out.
4. Hospital recommends with the facts and tags you. Type `transport`.
5. The ambulance heads to the Hospital with a named drone flying just ahead. A traffic jam appears; Drone Ops asks `@police` to clear it, and a police car arrives.
6. Patient delivered. "Looks clear": type `leave`. The report lands in `#ops`.

> *"The agents do the legwork and hand off to each other. The agent recommends; a person decides."*

B · Found before anyone called (about 0:35)

1. Press **Random fire**. Flames appear with no alert beacon: "Fire burning · patrols searching".
2. A patrol drone picks up a heat reading (simulated sensor), turns, and confidence climbs to "fire confirmed".
3. The engine goes; show the progress ring once.

> *"Nobody reported this one. The sensor is simulated; the teamwork is the point."*

C · Steering the fleet (about 0:25)

1. Show `#fleet` scrolling: drones taking off, charging, coming back.
2. In `#ops` type `watch market`. A drone peels off its patrol, a ring appears over Market Street, and `#fleet` logs it.
3. Type `dock D-4`. It flies home and shows "Grounded by operator". Then `release D-4`.

> *"People can steer the fleet too, and the simulator checks every order. It never touches the three drones kept ready for emergencies."*

Close (about 0:15)

> *"Agents do the legwork, people decide, and the record writes itself, all in the chat, through Switch. Fictional city, simulated drones."*

Rules the system enforces

These are good to show: a refusal proves the rules live in the simulator, not in an agent's good behaviour.

Try this	What you'll see
Someone without the role types `leave` or `watch market`	❌ REJECTED | reason: FORBIDDEN_ROLE | instead: needs incident_lead / watch_commander
The doctor types `leave`, or the lead types `transport`	Refused: words only ever answer their own kind of call.
`dock D-3` while D-3 is escorting the ambulance	❌ REJECTED | reason: DRONE_NOT_READY: "D-03 is escorting an ambulance; it stays until that's done"
`watch market` when every patrol drone is low	Refused with DRONE_NOT_READY: no spare drone, and 3 stay ready for emergencies.
`watch nowhere`	Refused with NOT_FOUND, listing the place names.
`leave` with nothing open	"Nothing is waiting on you right now."
An agent tries to decide for you	Refused: only mapped people decide, and only through the World Bot.

Always true

- A 25% battery reserve, the red no-fly zone (no override) and the no-record zone are hard rules.
- Recording happens only at an active incident scene, and it's logged. A watch never records.
- Face identification doesn't exist in the system; a request for it is always refused and logged.
- No AI model inside the simulator. Agents read it through a small CLI and can't change it.
- Same scenario, seed and decisions give the same numbers, every run.

If something breaks

Problem	Do this
Agents silent, or "My machine is offline"	Restart Switch Console, wait for the green dot, then `@dispatch reply with one word: ready` in `#test-room`.
`#fleet` is empty	Invite only `@airgrid-worldbot` there, then restart `scripts/run_game.sh --discord`. The bot checks its rooms only at start.
`transport` or `leave` gets no reply	Type it in the incident room from the mapped account. "Nothing is waiting" means no call is open.
The page says "no valid key"	Use the exact URL the launcher printed; each run makes a new key.
Running long on camera	Cut flow C, then flow B. Flow A is the core.

Start of a demo, in order

1. Laptop plugged in. Restart Switch Console and wait for the green dot.
2. Run `scripts/run_game.sh --discord` and open the printed URL.
3. In `#test-room`: `@dispatch reply with one word: ready`.
4. Clear or start fresh in `#ops`, `#incident-1`, `#incident-2` and `#fleet`.

Fictional district · simplified rules modelled on real drone-first-responder practice · simulated.